Sabrina DeSoto, Ph.D.

Computational astrophysicist with six years of research and coding experience — adept in project management, distilling complex information and communicating technical processes. A motivated self-starter eager to apply my expertise and unique perspective to address high-impact problems.

Denver, Colorado | sab.desoto@gmail.com
sablol.github.io/Sabrina-DeSoto/ | 303-913-4286 | Sabrina DeSoto

SKILLS

- Project ownership/management
- Conducting primary research
- Proposal Development
- Sourcing project funding
- Technical writing
- Programming/code development
- Computational modeling
- Data verification & visualization
- Analytical & statistical analysis
- Problem solving
- Interdisciplinary adaptability
- Leadership & collaboration
- Effective oral communication
- Task prioritization
- Teaching & mentoring

EXPERIENCE

RESEARCH FELLOW

Oak Ridge National Laboratory | Jan – Jun 2025 (6 months DOE grant-funded) | Knoxville, TN

- Designed/executed research plan, which **tripled analytical models (1→3) and produced 6 new datasets**
- Applied research expertise to improve realism in the massively parallel, simulation model: FLASH-X
- Led analytical analysis on simulation to quantify model performance and understand system behavior
- Built Python analytics pipelines to process, visualize and recognize trends in MB–TB scale datasets
- Worked with production grade deployment team to ensure code usability and publish work (in progress)

RESEARCH ASSISTANT

University of Denver | Aug 2019 – Oct 2025 (6 years) | Denver, CO

- Developed scalable Python functions to analyze largest dataset in field, **expanding data coverage by 4X**
- Discovered model issues via exploratory data analysis — created comparable toy model to quantify impact
- Established multi-institution research collaborations and led research between cross-disciplinary teams
- Mentored diverse research team in advanced computing, applied mathematics and data analysis
- Authored publications (DOI: 10.3847/1538-4357/ae0e6f, 10.3847/2041-8213/adbb63)

TEACHING ASSISTANT

University of Denver | Aug 2019 – Oct 2024 (4 total academic years) | Denver, CO

- **Improved student lab performance by 18%; achieved 98% positive instructor ratings**
- Communicated complex concepts to non-technical audiences, adapting explanations based on needs
- Guided students through structured problem-solving, experimental reasoning, and data interpretation
- Developed/improved instructional content based on performance, feedback, and real-time troubleshooting

EDUCATION

Ph.D. Physics

University of Denver | 2019 – 2025

- Dissertation Focus: Computational Astrophysics

B.A.Sc. Liberal Arts and Sciences

Quest University Canada | 2012 – 2016

- Thesis Focus: Physics

TECHNICAL PROFICIENCIES

- Expertise: Python, LaTeX, Bash, GitHub, MS Office (Word, PowerPoint, Excel, Outlook), Slack
- Familiar: High-Performance Computing, Fortran, VisIt, Globus, Mathematica, ML, LLM training

AWARDS

Graduate Student Research
US DOE Office of Science - 2025

Graduate Student Service
University of Denver - 2023

Teaching Assistant of the Year
University of Denver - 2020

OUTREACH

- Founded/managed Student Physics and Astronomy Club for Everyone (SPACE) at local middle school.
- Mentored DU SciTech campers and contributed to publication (DOI: 10.1186/s44329-025-00030-w)
- Advised students in the Academic Mentors Project, NDiSTEM conference and AAS conference